

MUHAMMAD HANAN

GIS Student | Remote Sensing | Spatial Analysis | Python

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PROFESSIONAL SUMMARY

GIS student in the 5th semester of a BS in GIS at the University of the Punjab, with hands-on experience in remote sensing, spatial analysis, and satellite image processing. Skilled in QGIS, ArcMap, and Google Earth Engine, applying Sentinel-2 and Landsat imagery for NDVI, NDBI, Land Surface Temperature, LULC classification, and Random Forest classification. Proficient in Python and Pandas, currently expanding into NumPy and GeoPandas. Seeking an internship in GIS, Remote Sensing, Geospatial Analytics, or GIS-focused Python to apply these skills and gain practical industry experience, with a long-term career direction in Machine Learning and GeoAI for Remote Sensing.

EDUCATION

University of the Punjab, Lahore 2024 – 2028
Bachelor of Science in GIS • Current Semester: 5th • CGPA: 3.49 / 4.00

TECHNICAL SKILLS

GIS & Spatial Analysis: Spatial Analysis, Spatial Data Processing, Vector & Raster Data, Georeferencing, Digitization, Shapefile Creation & Management, Coordinate Reference Systems (CRS), Cartography & Map Design, Thematic Mapping

Remote Sensing: Satellite Image Processing, Sentinel-2, Landsat, NDVI, NDBI, Land Surface Temperature (LST), LULC Classification, Supervised & Unsupervised Classification, Random Forest Classification

Python & Data Analysis: Python, Pandas (Current) | NumPy, GeoPandas (Currently Learning)

GIS Software & Platforms: QGIS, ArcMap, Google Earth Engine (GEE), Google Earth Pro, Microsoft Excel, Word, PowerPoint

Geospatial Data: Digital Elevation Models (DEM), OpenStreetMap, Shapefiles, Raster & Vector Data, NetCDF

SELECTED PROJECTS

Lahore Urban Heat & Vegetation Analysis *Google Earth Engine, QGIS, Sentinel-2, Landsat*

- Analyzed vegetation distribution, built-up areas, and land surface temperature across Lahore using satellite imagery.
- Generated NDVI from Sentinel-2 and Land Surface Temperature (LST) from Landsat; calculated NDBI to assess urbanization patterns.
- Integrated NDVI, LST, and NDBI to identify an extreme urban heat hotspot of approximately 197.85 km²; produced analytical maps in QGIS.

Punjab Land Use/Land Cover Classification *Google Earth Engine, Sentinel-2, Random Forest*

- Developed a supervised LULC classification workflow using Sentinel-2 multispectral imagery and Random Forest classification.
- Classified land cover into Trees, Water, Built-up, Bareland, and Grassland; generated classified maps for spatial interpretation.

Islamabad Land Use/Land Cover Classification *Google Earth Engine, Sentinel-2, QGIS, Random Forest*

- Processed Sentinel-2 imagery for supervised land-cover classification using Random Forest, incorporating NDVI, MNDWI, and built-up indicators.
- Generated and analyzed spatial LULC outputs; prepared maps for interpretation of land-cover distribution.

Pakistan Elevation & Terrain Mapping *ArcMap, Digital Elevation Model (DEM)*

- Created a Pakistan elevation map using DEM data, classified into six elevation classes, with hillshade for terrain visualization.
- Integrated geographic boundaries and cartographic elements into a professional final map layout for academic presentation.

CURRENT LEARNING & CAREER DIRECTION

- Python for GIS and geospatial data analysis; Python-based spatial data processing.
- NumPy — Currently Learning | GeoPandas — Currently Learning
- Machine Learning for Remote Sensing — Future Learning Direction
- GeoAI — Long-Term Career Specialization

LANGUAGES & REFERENCES

Languages: English, Urdu, Punjabi *References available upon request.*